

Topic quiz: Work and energy

IBDP Physics SL · Grade 11 · A.3 Work and energy · 10 multiple-choice questions

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Name: _____ Class: _____ Date: _____

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use $g = 9.81 \text{ m s}^{-2}$ and show any working next to the question.

Check yourself online: the same quiz with instant feedback and explanations — <https://biba0112.github.io/physics-sims/quiz-work-energy/>

[illegible]

Question 1

A child pulls a sled 10.0 m along flat snow with a force of 50.0 N at 30° above the horizontal. How much work does the pulling force do?

- (A) 43.3 J (B) 250 J
(C) 433 J (D) 500 J

Question 2

A box is pulled 5.0 m along a horizontal floor. How much work does the normal force do on the box?

- ☐ (A) Zero, because the normal force is perpendicular to the displacement
☐ (B) Zero, because the normal force is cancelled by the weight
☐ (C) $mg \times 5.0 \text{ m}$
☐ (D) A negative amount, because it opposes the motion

Question 7

A 0.500 kg Smart Cart rolls down a ramp. It loses 0.513 J of gravitational potential energy and gains 0.483 J of kinetic energy. How much energy is transferred to thermal energy by friction?

- ☐ (A) 0 J, because energy is conserved
- ☐ (B) 0.030 J
- ☐ (C) 0.483 J
- ☐ (D) 0.996 J

Question 8

A heavy pendulum is released from rest touching a student's nose. The student keeps perfectly still. Why can the pendulum not hit the student when it swings back?

- ☐ (A) It cannot rise higher than where it started, because its total energy cannot increase
- ☐ (B) It loses all its energy at the far end of the swing
- ☐ (C) Its mass is too large for it to come back
- ☐ (D) Gravity is weaker on the way back

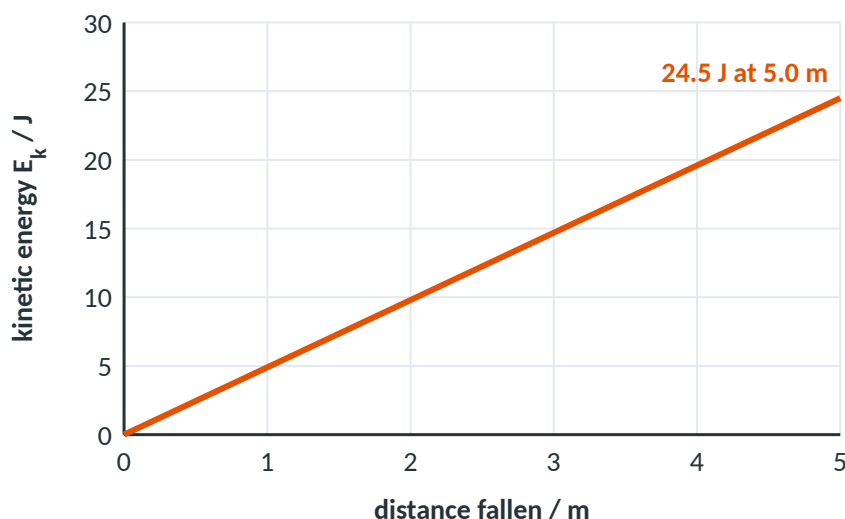
Question 9

A spring ($k = 200 \text{ N m}^{-1}$) is compressed by 0.050 m and launches a 0.020 kg toy straight up. Ignoring losses, how high does the toy rise?

- ☐ (A) 0.64 m
- ☐ (B) 1.27 m
- ☐ (C) 2.55 m
- ☐ (D) 51.0 m

Question 10

A 0.50 kg ball is dropped. The graph shows its kinetic energy against the distance fallen. What does the gradient of the graph represent?



- ☐ (A) The ball's speed, 4.9 m s^{-1}
- ☐ (B) The ball's mass, 0.50 kg
- ☐ (C) The acceleration of free fall, 9.81 m s^{-2}
- ☐ (D) The ball's weight, 4.9 N